1. System Targets

Target Output Data Rate	1 ksps
Oversampling Ratio (OSR)	8
Target Sampling Rate	8 ksps

2. Clock Cycle Allocation

Conversions per output code	8 conversions
Cycles per A/D conversion	9 clock cycles
Padding for Noise Shaping (NS)	3 clock cycles (average)
Total cycles per conversion	12 clock cycles (average)
Total cycles per output code	96 clock cycles

3. Clock Frequency Requirements

Required Clock Rate	96 kHz (Period ≈ 10.417 µs)
Relaxed System Clock Input	100 kHz (Period = 10.0 µs)
Resulting Output Data Rate at 100 kHz	~1.042 ksps

> ***Note:*** *The system is designed for a nominal 96 kHz input clock, meanwhile relaxing the strict 96 kHz requirement to accommodate a 4.16% higher upper target capability of 100 kHz.*

4. Timing Diagrams

4.1 Timing per Code Output

```
/CONV0\/CONV1\/CONV2\/CONV3\/CONV4\/CONV5\/CONV6\/CONV7\
\     /\     /\     /\     /\     /\     /\     /\     /
```

** CONV: Conversion (single 8-bit SAR resolution + NS loop)*

4.2 Timing per Conversion (Non-Overlapping)

```
/SAMP\/BIT7\/BIT6\/BIT5\/BIT4\/BIT3\/BIT2\/BIT1\/BIT0\/ NS \/ NS \/ NS \
\    /\    /\    /\    /\    /\    /\    /\    /\    /\    /\    /\    /
```

** SAMP: Sample phase | BITx: Bit decision phase | NS: Noise shaping loop phase*

4.3 Timing per Conversion (Overlapped Architecture)

An overlapped architecture variant can be considered to optimize timing:

```
/SAMP\/BIT7\/BIT6\/BIT5\/BIT4\/BIT3\/BIT2\/BIT1\/BIT0\/FIN \/SET \/    \
\    /\    /\    /\    /\    /\ NS /\ NS /\ NS /\ NS /\    /\    /\    /
```

** FIN: Final residue transfer into the integrator | SET: Integrator settling phase*

5. Asynchronous Design Notes

Non-overlapping clocks will be utilized for robust asynchronous event handling.